

ESMI Applied Mathematics Project Syllabus

Queueing Theory

5 June – 4 August 2023

Instructor: Laurel Newman (lan66@cornell.edu)

Lecture Location: 104 Carpenter Hall

Lecture Time:

- Weeks 1 & 2: 10:00-12:00 M\T\W\R\F
- Weeks 3-9: 1:00-3:00 M\W\F

Office Hours: TBD

1 Introduction

This summer we will gain experience in doing mathematical research and cover a brief introduction to queueing theory. We will cover probability, some relevant statistical distributions, and Python. Students will work in small research groups to develop their own question about a real-world queueing system and use mathematical and computational tools to gain insight into the system. Students will write up their results and present them at the end of the program.

1.1 Resources

Here are some additional resources you may wish to make use of if you wish to explore a topic in more depth than we cover in lecture.

Some further reading on Queueing Theory: https://www.cs.ucf.edu/~lboloni/Teaching/EEL6785_Fall2010/slides/QueueingTheory.pdf

NumPy for beginners: https://numpy.org/doc/stable/user/absolute_beginners.html

Introductory probability: <https://math.dartmouth.edu/~prob/prob/prob.pdf>

A great resource for brainstorming queueing problems: <https://opexlearning.com/resources/queueing-theory/>

1.2 Timeline

1. Monday 5 June: program starts, beginning first two weeks of daily lectures

2. Tuesday 20 June: students' summer courses begin, we now meet M/W/F.
 - Mondays: skills and best practices
 - Wednesdays: continuing lectures/work days
 - Fridays: work days
3. Friday 23 June: students finalize their research groups and submit their project proposals and group contracts
4. Friday 21 July: students submit their draft research report, poster, and presentation
5. Friday 28 July: "working draft" of final research report, poster and presentation due
6. Wednesday 2 August: students submit their final research report, poster, and presentation
7. Thursday 3 August: Summer Research Symposium

1.3 Schedule

Week	Day	Lecture
1	5 June	Introduction
	6 June	Intro to Queues
	7 June	Intro to Prob
	8 June	Prob II: Independence
	9 June	Prob III: Useful distributions
2	12 June	Queues with Randomness
	13 June	Intro to Python
	14 June	Python II
	15 June	NumPy for Queuing Theory
	16 June	Intro to Research
3	19 June	Juneteenth
	21 June	Graphing your results
	23 June	Project Proposals & Group Contracts Due
4	26 June	Coding Documentation
	28 June	Project Planning
	30 June	Work Day
5	3 July	How to write up your results
	5 July	
	7 July	Work Day
6	10 July	Effective Math Communication
	12 July	
	14 July	Work Day
7	17 July	
	19 July	
	21 July	Poster, presentation, report draft due
8	24 July	
	26 July	
	28 July	“Working draft” of poster, presentation, report due
9	31 July	Presentation Prep
	2 August	Presentation Prep & final deliverables due
	4 August	Presentations